# Kolmogorov–Arnold Networks for Higgs Boson Classification

A comparison with gradient-boosted trees on the FAIR Universe dataset

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**First version** (preliminary report; final version to follow at the end of the internship)

**Abstract**

We study Kolmogorov–Arnold Networks (KAN) for the classification of Higgs boson events ($H \to \tau\tau$) against background on the FAIR Universe HiggsML dataset (a public sample of 2,000,000 events), benchmarked against XGBoost. Models are evaluated with the weighted AUC and, as the primary physics metric, the Approximate Median Significance (AMS) normalised to a $10\,\mathrm{fb}^{-1}$ luminosity. A properly tuned deep KAN outperforms XGBoost on both AUC (0.882 vs. 0.853) and AMS (3.69 vs. 2.27). An adversarially decorrelated KAN is about $9\times$ more robust to energy-scale systematics while retaining its accuracy advantage. This is a preliminary report; further analyses are deferred to the final version.

## 1 Introduction

Machine-learning classifiers are central to modern high-energy-physics (HEP) analyses: almost every measurement or search relies on separating a signal process from large backgrounds, and the quality of that separation directly limits the physics reach. Classifiers that are simultaneously accurate, robust to systematic uncertainties, and interpretable are therefore of broad and growing interest.

Kolmogorov–Arnold Networks (KAN) are a recently proposed neural architecture that places learnable univariate functions on the network edges, offering high expressivity together with interpretability — one can inspect *why* a decision is made, which is valuable in physics. Gradient-boosted decision trees (XGBoost) are the de facto standard for tabular HEP data and are typically hard to beat with neural networks, providing a demanding baseline. To assess whether KAN is competitive we require a representative and reproducible benchmark.

We therefore study KAN on a typical, well-understood task: separating the Higgs-boson decay $H \to \tau^+\tau^-$ from its backgrounds (notably $Z \to \tau\tau$, $t\bar{t}$ and diboson production), a channel for which a large open dataset — the FAIR Universe HiggsML sample — is available. This is a strongly class-imbalanced problem in which the relevant figure of merit is the *statistical significance* of the extracted signal rather than raw accuracy. The objective of this internship is to implement KAN from scratch, evaluate it fairly against XGBoost, and assess its accuracy (AUC), physics significance (AMS), robustness to systematic uncertainties, and interpretability. We find that, once properly tuned, KAN meets all of these criteria and surpasses the XGBoost baseline.

## 2 Related Work

KANs were introduced by Liu et al. [1, 2] as an alternative to multilayer perceptrons, replacing fixed node activations with learnable spline functions on the edges. For tabular data, boosted-tree ensembles such as XGBoost [3] remain very strong baselines. The FAIR Universe HiggsML challenge [6] provides a large simulated dataset designed to study robustness to systematic uncertainties; the AMS significance used here follows the asymptotic formulae of Cowan et al. [4]. Robustness to nuisance parameters is addressed via adversarial "pivoting" as proposed by Louppe et al. [5]. Preliminary explorations of KANs on this dataset motivated the present, full-scale and carefully tuned comparison.

## 3 Methods

**Dataset.** We use the "blackSwan" sample of the FAIR Universe HiggsML public data, comprising 2,000,000 simulated $pp$ collisions with 28 features each. We adopt this sample because it is large enough to train and evaluate the models robustly while remaining computationally tractable; it is a subset of the substantially larger full simulated production. The data are split in a stratified manner into 1.4M training, 300k validation and 300k test events; missing values are imputed by feature means and all inputs are standardised using training-set statistics.

**Metrics.** We report the weighted AUC and, as the primary metric, the AMS,

$$\mathrm{AMS} = \sqrt{2\left[(s+b)\,\ln\!\left(1+\tfrac{s}{b}\right) - s\right]}, \tag{1}$$

where $s$ and $b$ are the summed signal and background weights passing a decision threshold, maximised over the threshold. Event weights are normalised so that the test set reproduces the expected counts of a $10\,\mathrm{fb}^{-1}$ run ($s = 1015$ signal, $b = 1{,}050{,}370$ background events, following the dataset's "LHC Events" reference).

**Models.** The XGBoost baseline uses 8587 trees of depth 2 at a learning rate of $9.1 \times 10^{-4}$. These hyperparameters were adopted from a prior hyperparameter search rather than re-optimised on this dataset; a dedicated XGBoost hyperparameter optimisation on the present data is planned for the final version. The KAN is implemented from scratch in PyTorch: each edge function is a B-spline evaluated via the Cox–de Boor recursion, with adaptive knot placement, optional coarse-to-fine *grid extension* ($G$=3 $\rightarrow$ $G$=10) and $L_1$ regularisation of the spline coefficients. The main model is a deep KAN with three layers of width 96, grid size $G$=5 and spline order $k$=3, giving about $1.9 \times 10^5$ trainable parameters; it is trained with Adam, a cosine learning-rate schedule, mixed precision and early stopping. An adversarial variant adds a decorrelation network to suppress sensitivity to the energy scale.

**Robustness protocol.** To emulate a mis-calibrated detector we scale the momentum- and energy-related features — the transverse momenta ($p_T$ of the lepton, hadronic tau and jets, and their sum `DER_sum_pt`), the reconstructed masses (`DER_mass_*`) and the missing transverse energy (`PRI_met`) — by a common factor $\gamma \in [0.8, 1.2]$, and report the maximal AUC drop from the nominal $\gamma = 1$ (lower is more robust).

## 4 Results and Discussion

Table 1 summarises the main results on the held-out test set (mean ± standard deviation over 200 bootstrap resamples).

Table 1: Performance on the held-out test set (300,000 events). AMS is at $10\,\mathrm{fb}^{-1}$ LHC luminosity; "Max drop" is the AUC degradation under energy-scale shifts (lower is more robust).

| Model | AUC | AMS | Max AUC drop |
|---|---|---|---|
| KAN (deep, $[96, 96, 96]$) | **0.8820** | 3.695 | 0.101 |
| KAN + grid extension | 0.8799 | **3.935** | 0.090 |
| KAN-Adversarial | 0.8743 | 2.949 | **0.006** |
| XGBoost (baseline) | 0.8533 | 2.269 | 0.052 |

**KAN outperforms XGBoost.** Once properly configured, the deep KAN exceeds XGBoost on both AUC (0.882 vs. 0.853) and the physics-primary AMS (3.69 vs. 2.27). Both models are trained and evaluated on the same data, with XGBoost using a fixed, standard configuration (see Section 3). The gain came primarily from architecture (depth, cubic splines and a higher learning rate); a shallow, low-learning-rate KAN underperforms. The ROC curves (Fig. 1) show the KAN lying above XGBoost across the operating range.

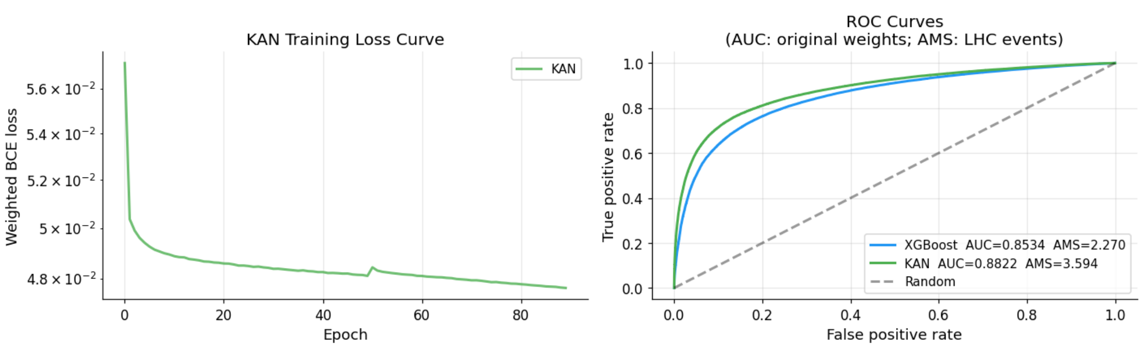


Figure 1: Left: KAN training loss converging. Right: ROC curves — the tuned KAN (AUC 0.882) lies above XGBoost (AUC 0.853).

**Robustness.** The high capacity of the deep KAN makes it more sensitive to the energy-scale systematic than XGBoost (Fig. 2). Adversarial decorrelation resolves this: the adversarial KAN reduces the maximal AUC drop to 0.006 ($\approx 9\times$ more robust than XGBoost) while still beating XGBoost on AUC and AMS, making it the best all-round model.

**Interpretability.** Because each KAN edge is an explicit function, feature importance and learned activations are directly inspectable (Fig. 3). KAN's most important feature is the visible di-tau invariant mass (`DER_mass_vis`), consistent with established HEP intuition. We stress that the KAN and XGBoost importances in Fig. 3 are computed by *different* measures (the $L_2$ norm of the spline weights and the XGBoost "gain", respectively) and are therefore not directly comparable; the two models also rely on rather different features. A common, model-agnostic measure (e.g. permutation importance) is planned for the final version.

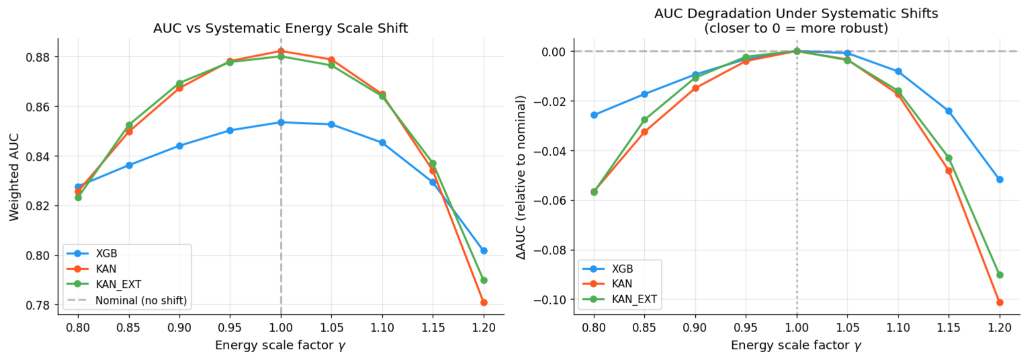


Figure 2: AUC under energy-scale shifts $\gamma \in [0.8, 1.2]$. The deep KAN is most accurate at $\gamma = 1$ but more sensitive away from it.

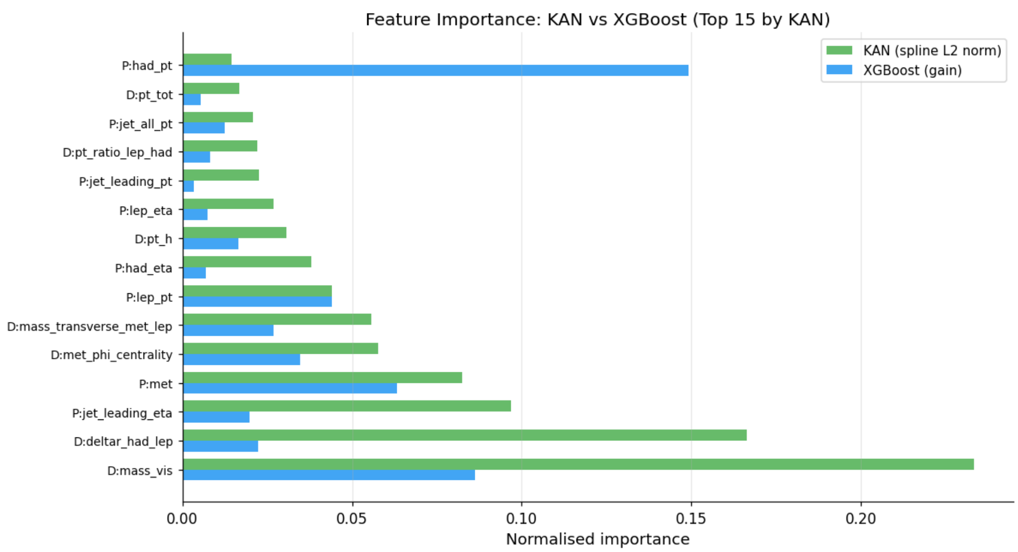


Figure 3: Feature importance for KAN and XGBoost. KAN's leading feature, `DER_mass_vis`, is physically meaningful.

# 5 Conclusion

A properly tuned KAN is a competitive and interpretable alternative to XGBoost for $H \to \tau\tau$ classification, outperforming it on both AUC and AMS on the held-out test set, with an adversarial variant offering substantially improved robustness to systematic uncertainties. The central practical finding is that architecture — network depth, spline order and learning rate — determines whether KAN matches or surpasses a strong tree baseline; the same dataset and a fixed XGBoost configuration were used throughout.

As this is a preliminary report, the following analyses are planned for the final version:

- a dedicated hyperparameter optimisation of the XGBoost baseline on this dataset;
- a study of performance versus the number of training events (learning curves);
- a study of the correlation between the XGBoost and KAN predictions — motivated by their rather different feature usage (Fig. 3) — together with a common, model-agnostic evaluation

of feature importance;

- a systematic hyperparameter and ablation study of the KAN (depth, grid size, spline order, learning rate) with $k$-fold cross-validation and fuller uncertainty quantification;

- additional baselines and a deeper physics interpretation of the learned spline functions.

## References


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[5] G. Louppe, M. Kagan and K. Cranmer, *Learning to Pivot with Adversarial Networks*, NeurIPS, 2017.

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